

BÀI TẬP NGÀY 14/4:

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Bài1. Tạo server chat giữa các client

The screenshot shows a WSL terminal window titled 'Lap_trinh_mang [WSL: Ubuntu-22.04]'. It contains three terminal panes. The left pane shows the server running on port 8888, waiting for connections. The middle pane shows a client (client 1) connecting and sending messages. The right pane shows another client (client 2) connecting and sending messages. The server logs show the connections and messages received from both clients.

```
chat_server.c Bai-tap-7-4 chat_server.c Bai-tap-14-4 U X BÀI TẬP NGÀY 7-4.pdf
Bai-tap-14-4 > C chat_server.c > Client
#include <stdio.h>

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
daochien@Daochien:~/Lap_trinh_mang$ cd Bai-tap-14-4
daochien@Daochien:~/Lap_trinh_mang/Bai-tap-14-4$ gcc -o chat_server chat_server.c
daochien@Daochien:~/Lap_trinh_mang/Bai-tap-14-4$ ./chat_server 8888

Chat Server running on port 8888
Waiting for connections...

New connection from 127.0.0.1:48454
New connection from 127.0.0.1:48468
Client authenticated: 1 (Chien)
Client authenticated: 2 (Quynh)
[1] Xin chao
[1] Ban khoe khong
[2] Toi khoe
[2] Con ban

daochien@Daochien:~/Lap_trinh_mang$ nc localhost 8888
host 8888
Please enter your identification in format: client_id: client_name
1: Chien
Welcome 1! You are now connected.
2026/04/20 20:03:06 *** 2 has joined the chat ***
Xin chao
Ban khoe khong
2026/04/20 20:03:24 2: Toi khoe
2026/04/20 20:03:27 2: Con ban

daochien@Daochien:~/Lap_trinh_mang$ nc localhost 8888
host 8888
Please enter your identification in format: client_id: client_name
2: Quynh
Welcome 2! You are now connected.
2026/04/20 20:03:12 1: Xin chao
2026/04/20 20:03:19 1: Ban khoe khong
Toi khoe
Con ban
```

- 2 client đăng ký id với server và gửi tin nhắn cho nhau thông qua id định danh

Bài2. Tạo một telnet_server kiểm tra thông tin đăng nhập người dùng gồm user và pass

The screenshot shows a VS Code terminal window with two panes. The left pane shows the server's execution, and the right pane shows the client's interaction.

Left Pane (Server):

```
daochien@Daochien:~/Lap_trinh_mang/Bai-tap-14-4$ gcc -o telnet_server telnet_server.c
daochien@Daochien:~/Lap_trinh_mang/Bai-tap-14-4$ ./telnet_server 8888
Loaded 3 users from users.txt

TELNET SERVER
Port: 8888
Users file: users.txt
Loaded 3 users

Server is running. Waiting for connections...
Use 'telnet localhost 8888' to connect
Or 'nc localhost 8888'

New connection from 127.0.0.1:48388 (fd=4)
Client unknown (fd=4) disconnected
New connection from 127.0.0.1:48012 (fd=4)
User admin authenticated (fd=4)
Executing: ls > /tmp/cmd_output_4.txt 2>&1
Executing: -la > /tmp/cmd_output_4.txt 2>&1
sh: 0: Illegal option -
Executing: pwd > /tmp/cmd_output_4.txt 2>&1
Executing: date > /tmp/cmd_output_4.txt 2>&1
Executing: free -m > /tmp/cmd_output_4.txt 2>&1
```

Right Pane (Client):

```
daochien@Daochien:~/Lap_trinh_mang$ nc localhost 8888

Welcome to Telnet Server
Please login with: username password

Login: admin admin

Welcome admin! Authentication successful.
You can now execute commands.
Type 'exit' or 'quit' to disconnect.

$ ls
chat_server
chat_server.c
telnet_server
telnet_server.c
user.txt
users.txt
$ -la
Error: Cannot execute command or read output
$ pwd
/home/daochien/Lap_trinh_mang/Bai-tap-14-4
$ date
Mon Apr 20 20:17:23 +07 2026
$ free -m
```

	available	total	used	free	shared	buff/cache
Mem:	2513	3751	1159	2373	3	218
Swap:		1024	0	1024		

- Client sau khi login thành công và thực hiện các lệnh ls, dir, pwd, ...